260114APPROACH

一、预处理策略

深度 h 用抛物面的形状，但是深度是从地形表面开始计算的，如下图 1 所示：

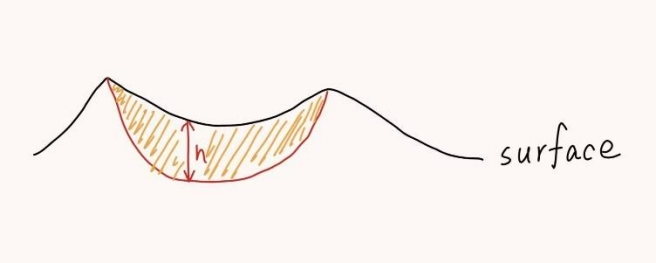

图 1 月海玄武岩的形状

主要月海的分布，如下图 2 所示：

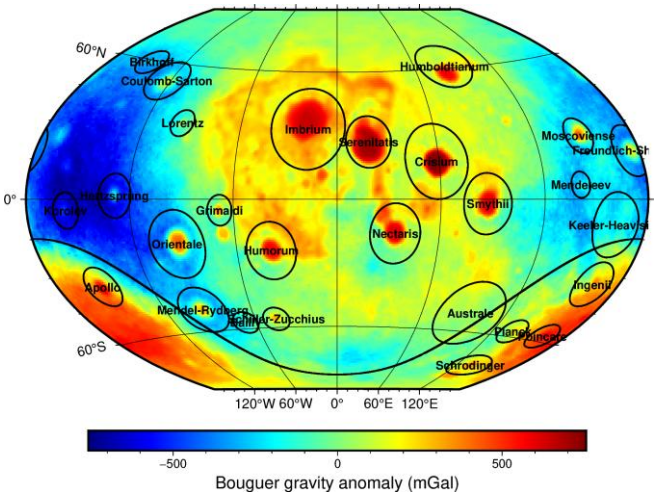

图 2 主要月海分布

产生的重力异常，如下图 3 所示：

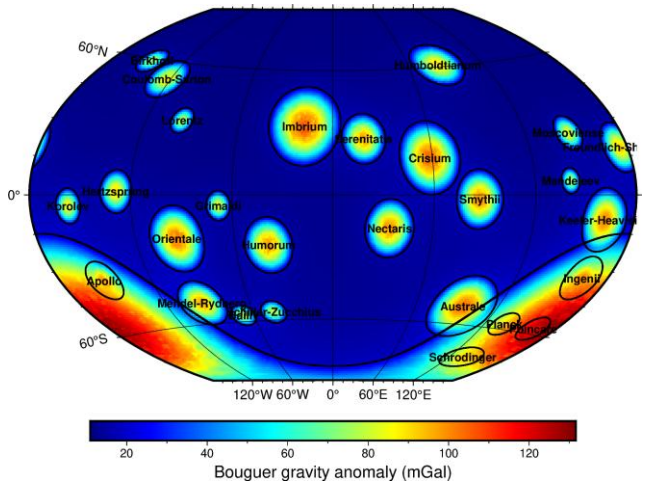

图 3 月海的玄武岩产生的重力异常

扣除月海玄武岩的布格重力异常如下图 4 所示：

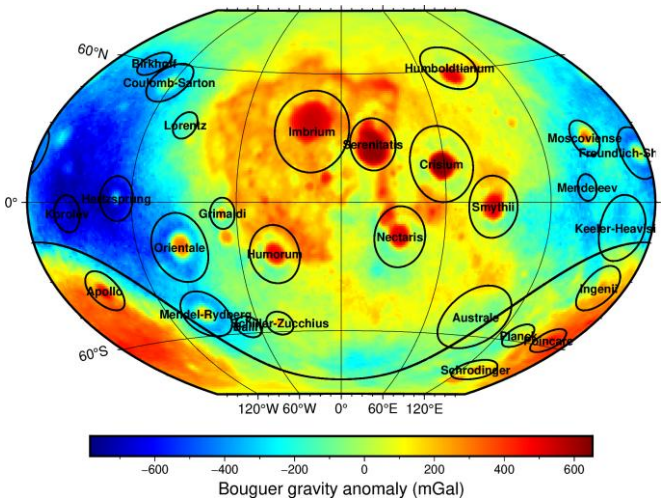

图 4 扣除月海玄武岩产生的布格重力异常

二、反演结果

反演结果，最佳正则化参数10^{-10}，最佳参考平面$40km$，最佳参考密度$450kg/m^3$，反演最终结果如下所示：

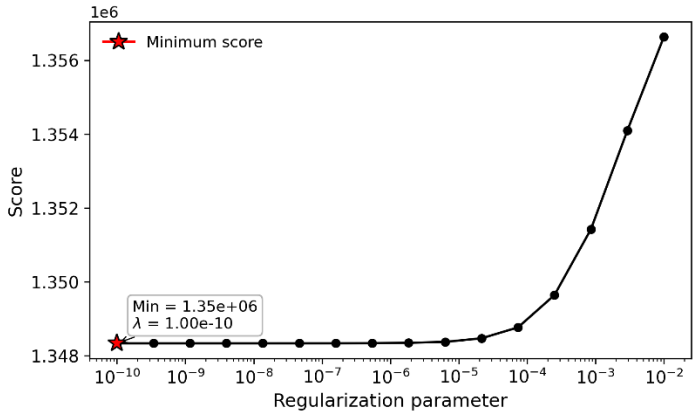

图 5 正则化参数

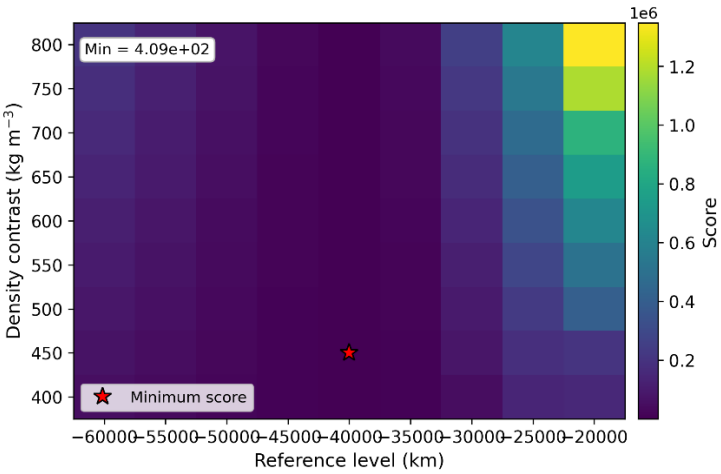

图 6 密度对比与参考平面

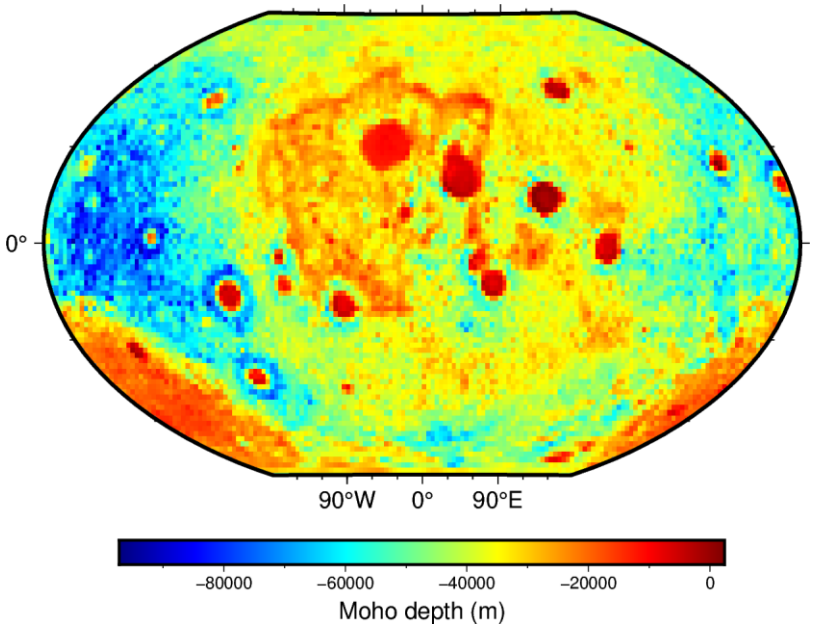

图 7 反演的月壳深度

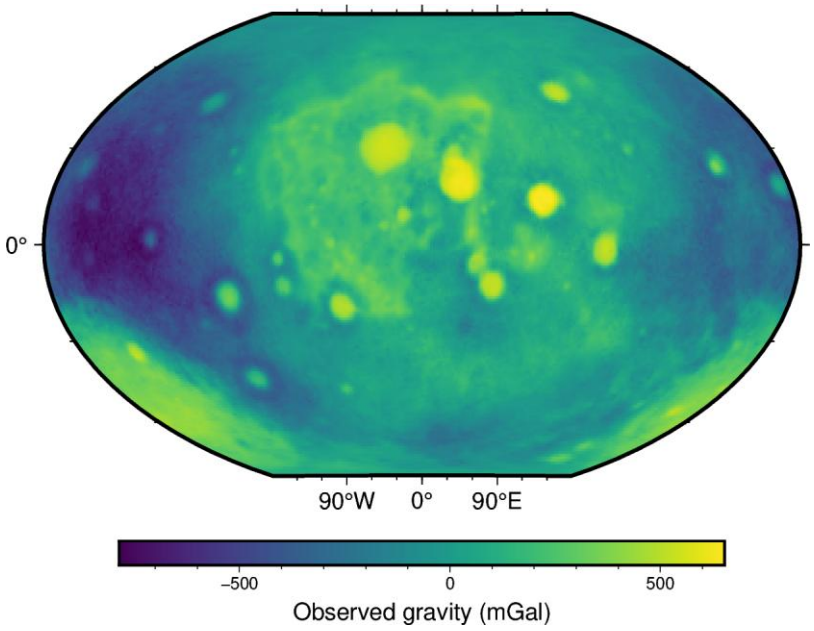

图 8 观测到的重力异常

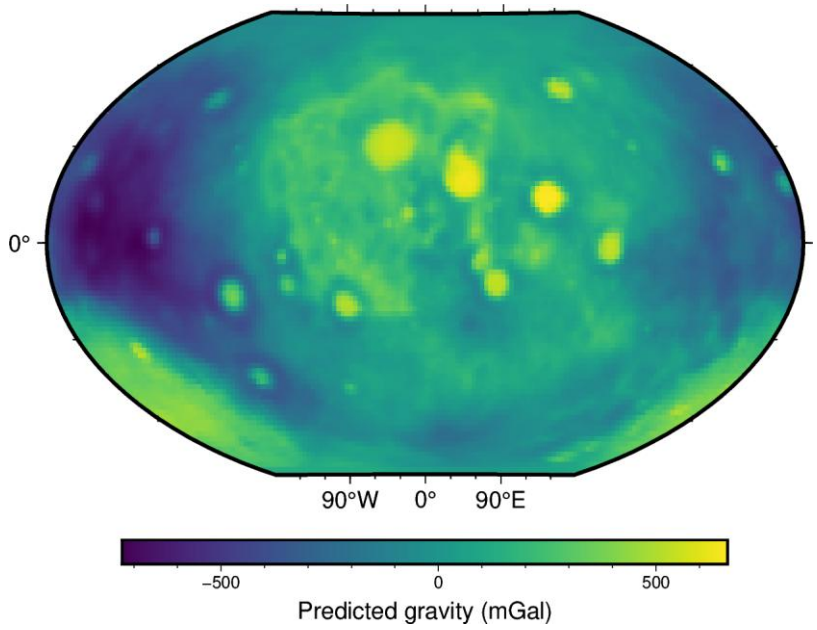

图 9 正演的重力异常

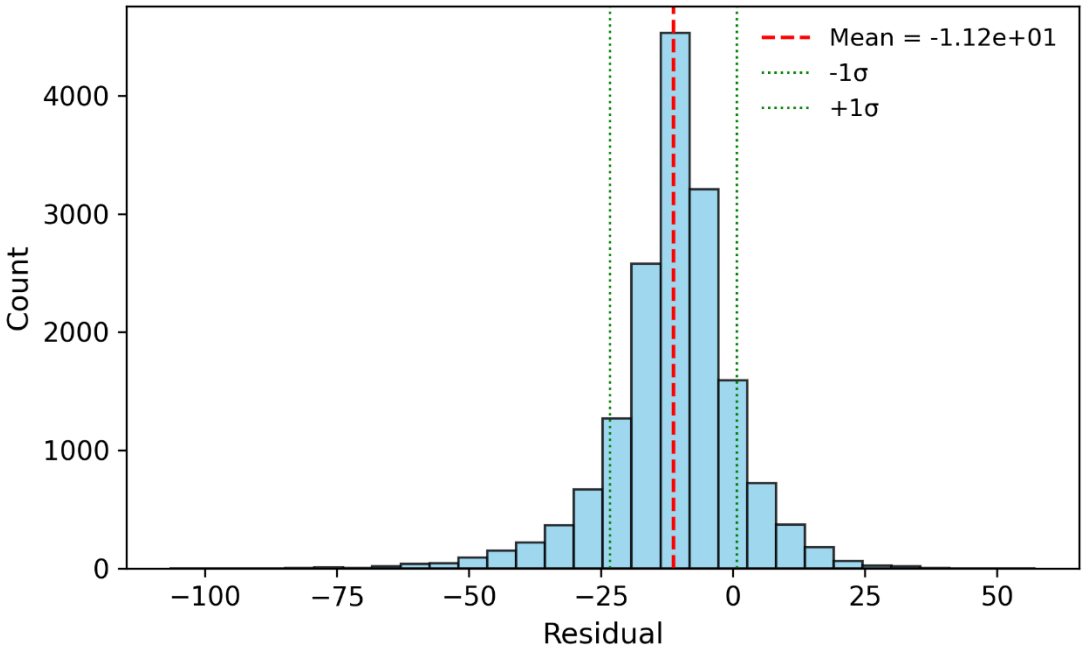

图 10 残差直方图